

Aayush

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SUMMARY

Machine Learning Engineer with 4+ years of experience developing and deploying ML systems across the full project lifecycle, from experimentation through to production. Currently building real-time speech localization systems at an AI startup, with hands-on work in TTS research, LLM fine-tuning, voice cloning, and model evaluation. Strong foundation in ML and data science, with experience across both research and applied settings.

EDUCATION

Purdue University, West Lafayette,

Master of Science, Electrical and Computer Engineering

Courses: Computer Vision, AI, Data Engineering, Reinforcement Learning, Probability & Statistics

Research: Geospatial flood risk modeling for infrastructure risk assessment and planning

USA

Dec 2024

Shiv Nadar University, Greater Noida,

Bachelor of Technology, Electrical and Electronics Engineering

Courses: Machine Learning, Big Data Analytics, Signal Processing, Probability & Statistics

Achievements: Dean's List; Graduated with Distinction; Full tuition waiver; IEEE publication

India

May 2019

SKILLS

ML & AI: PyTorch, Hugging Face, Scikit-Learn, XGBoost, Pandas, NumPy, LangChain, FAISS

Programming: Python, SQL, FastAPI, REST APIs, Docker, MLflow, Git, Linux

Data & Cloud: AWS, GCP, BigQuery, PySpark, Kestra, Airflow, Terraform, Tableau

EXPERIENCE

AI Localization Startup (Stealth)

ML Engineer

USA

Mar 2026 – Present

- Contributing to a real-time multilingual speech localization system, with a focus on TTS research and model evaluation. Built an evaluation framework to benchmark synthesis quality across 5+ model variants using MOS, WER, and speaker similarity, helping the team surface performance gaps and make faster model selection decisions.
- Fine-tuned Gemma on low-resource language data to improve text normalization ahead of speech synthesis, and ran structured evaluations of voice cloning models across target languages to identify failure patterns and inform the production pipeline design.

BME

Data Scientist

USA

Mar 2025 – Feb 2026

- Ran cohort analysis, A/B tests, and participant segmentation on program engagement data to surface patterns that fed directly into planning and resource allocation decisions for program leadership, supporting operations across a network of 11,400+ participants.
- Built the data pipelines and REST APIs that automated recurring reporting across locations, cutting roughly 15 hours of manual work per week and giving program teams real-time visibility into operational performance.

ABB Solar R&D

Software Engineer

India

Jan 2019 – Apr 2021

- Contributed to a cloud-connected remote monitoring platform that gave solar asset operators real-time visibility across 100+ PV plants and roughly 50,000 connected devices, with live telemetry streaming to Azure and supporting oversight of approximately 30 GW of installed capacity.
- Built the data ingestion and ETL layer collecting tens of GBs of daily telemetry from distributed plant sites, with schemas designed to serve both live operational dashboards and long-term historical analysis.
- Applied ML models and statistical methods to operational sensor data for fault detection, performance trend analysis, and predictive maintenance. Used approaches such as tree-based models, regression analysis, and clustering to help operators identify equipment issues early, understand performance patterns, and prioritize maintenance decisions.

PROJECTS

TV Series Intelligence — NLP and LLM Fine-Tuning

- Analyzed subtitle data from a popular TV series to extract themes using a BART zero-shot classifier and built an interactive character relationship network with named entity recognition and NetworkX. Fine-tuned LLaMA 3.1-8B using LoRA and 4-bit quantization to create a conversational agent that authentically captures the character's personality and conversational style.

AI Legal Assistant — RAG-Powered Document QA

- A retrieval-augmented generation application for querying large legal document collections in plain language, built on LangChain, FAISS, and AWS Bedrock with LLaMA 3. Added an LLM-as-a-judge evaluation layer using Evidently and response guardrails after observing inconsistent factual grounding in early outputs, which measurably improved accuracy across test query sets.

NY Taxi Ride Analysis — ML Pipeline at Scale

- Built an automated ELT pipeline using Kestra to process over 1 billion NYC taxi records into BigQuery, capable of incremental loading and data quality checks. Trained and deployed an XGBoost fare prediction model in Docker. Analyzed demand distributions, pricing anomalies, and ride patterns across boroughs and time periods.

CERTIFICATIONS

AWS Certified Data Engineer (DEA-C01, 2025) • Agentic AI – DeepLearning.AI • VLM Bootcamp – OpenCV University